RISK 231: Introduction

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My Introduction

- Professional Intro
  - LinkedIn Profile
  - Publication List
  - Business Administration/Mathematics Major at Seoul National University
  - Assistant Manager in DB Insurance (South Korea)
- Personal Intro
  - (Used to) play the guitar
  - Love to eat, cook to eat, work out to eat

Required Materials

- Note that you can download Office 365 for free as an SFU student.
- R and RStudio are also free.
- If possible, I would encourage you to bring your own laptop so that you can practise along with the lectures during class time.

Why do actuaries need to know tools for data-driven decision making?

What does an actuary do?

Candy bar companies sell candy bars!

- **PRICE = COSTS + PROFIT**
- What are the costs to include when deciding on the price of a candy bar?

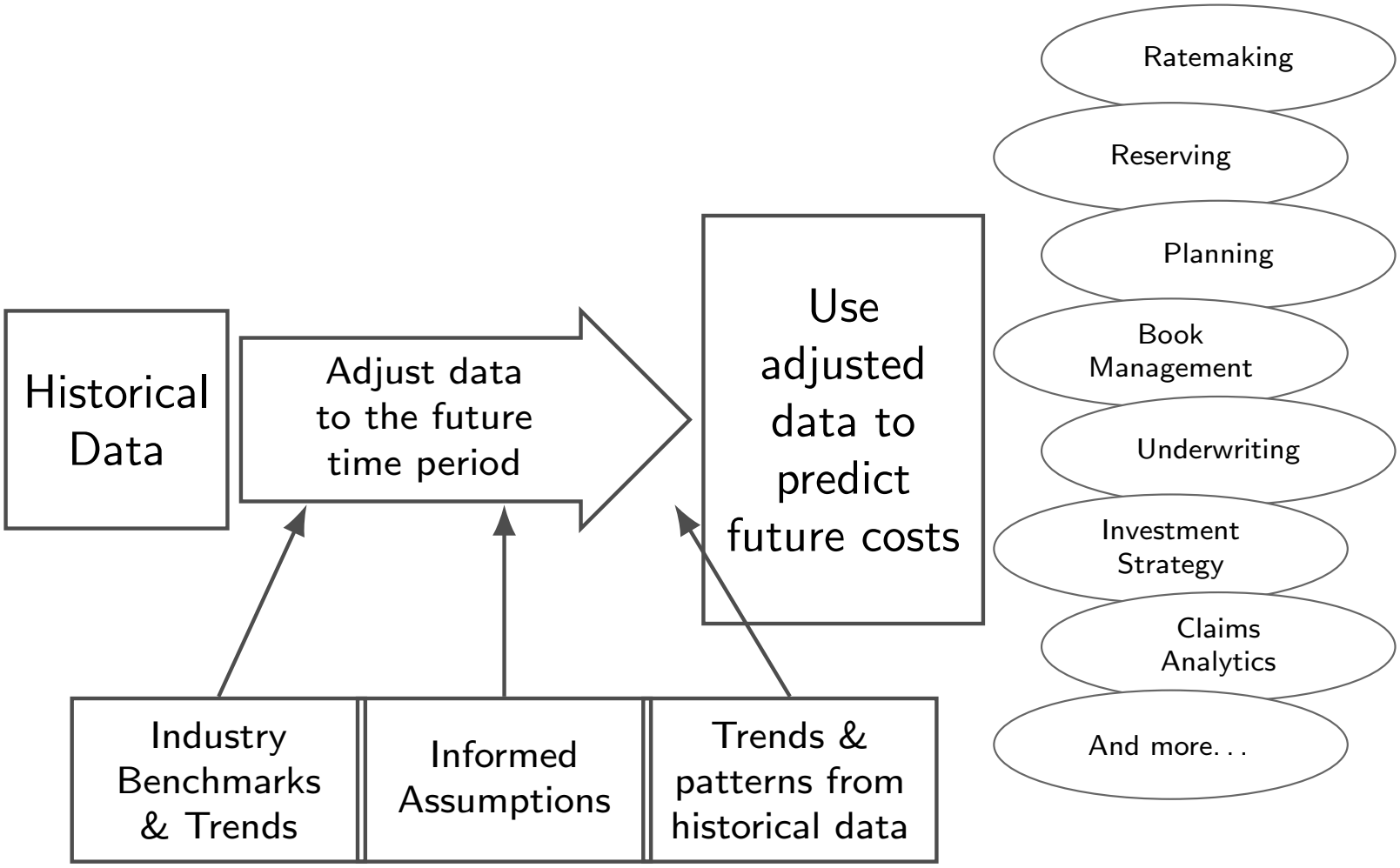
What do insurance companies sell?

- A promise to indemnify the policyholder for **future** costs.
- How do we price a product where all the costs are in the future?!?
- PRICE = COSTS + PROFIT
- **PREMIUM = EXPECTED FUTURE LOSSES + EXPECTED FUTURE EXPENSES + DESIRED FUTURE PROFIT**

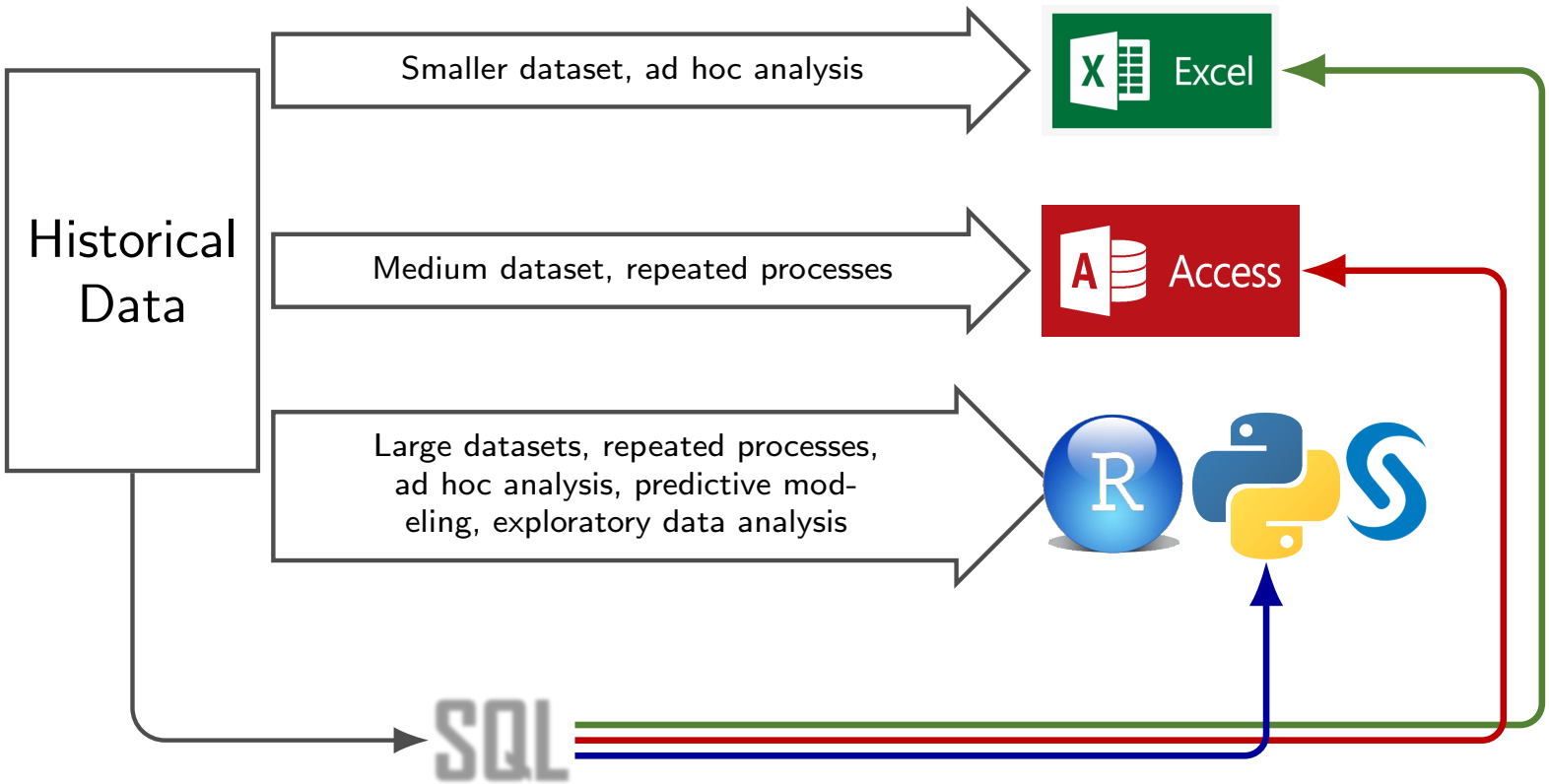
What does an actuary do?



Why do actuaries need to know programming?



Why do actuaries need to know tools for data-driven decision making?



Some remarks

- While the course has been designed primarily to equip actuarial students with the computational skills they need, these skills are in fact versatile and can be used in any profession with office work.
- Note that MS Access is not available with SFU's Office 365 license, so it is not covered in this class – in my understanding, MS Access is a version of a SQL application with better visualization.
- I think the most important reason that you need to learn Excel is. . .

To avoid unnecessary overtime work and sneak out from your workplace on time!

Suggestion for actuarial students

- I would strongly suggest that you
  - pass SOA Exam P (STAT 270 + α) – costs USD 260,
  - pass SOA Exam FM (ACMA 201) – costs USD 260,
  - and find a co-op position as an actuarial student.
- After you have finished your first co-op, you can decide whether you would like to stay in the profession or look for another career. (Anyway, having one co-op opportunity does not hurt at all!)
- It is known that without one or two actuarial exams, it is not easy to secure an actuarial co-op (two is preferable).